



TEAM NAME:

Off by One

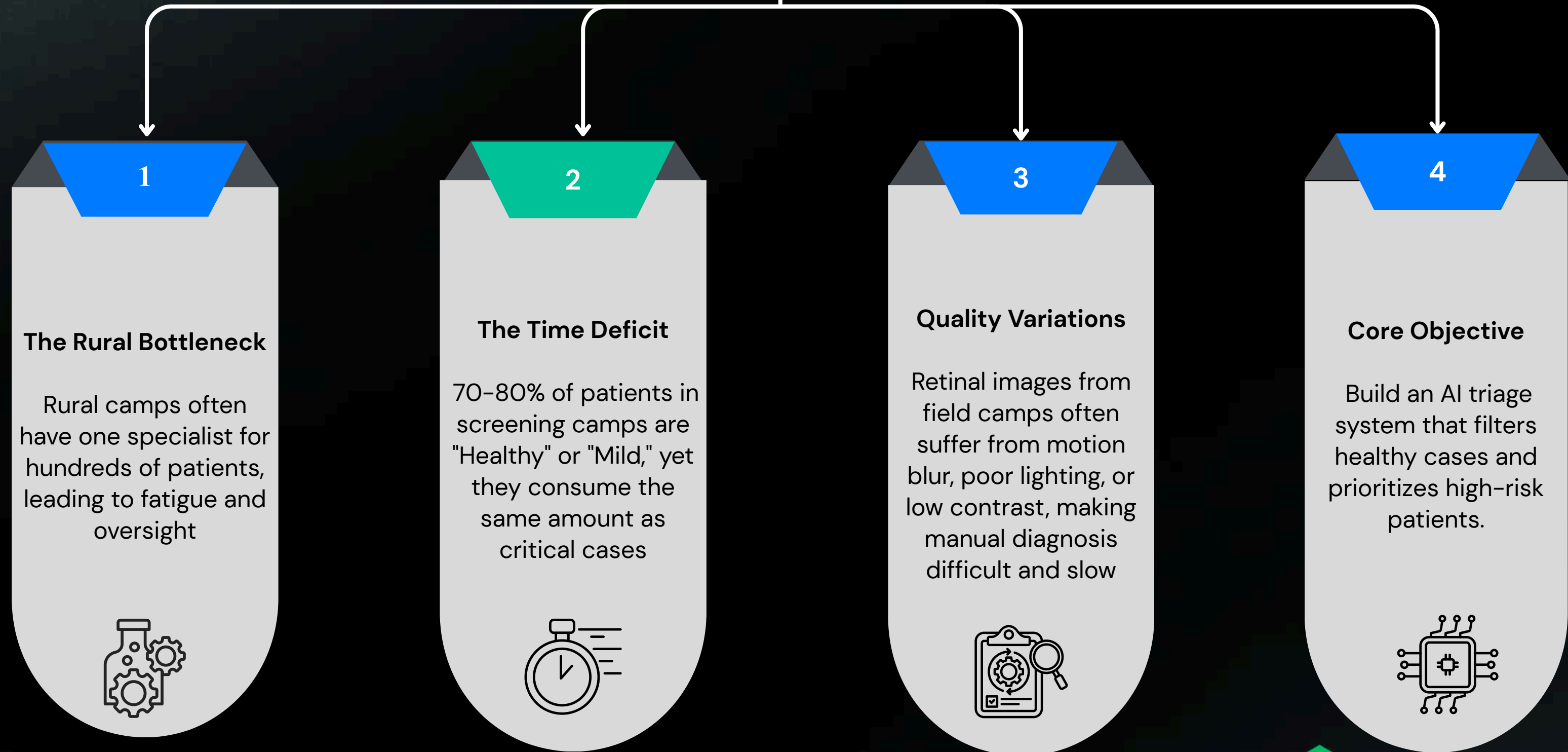
COLLEGE NAME:

Sri Venkateswara College of Engineering

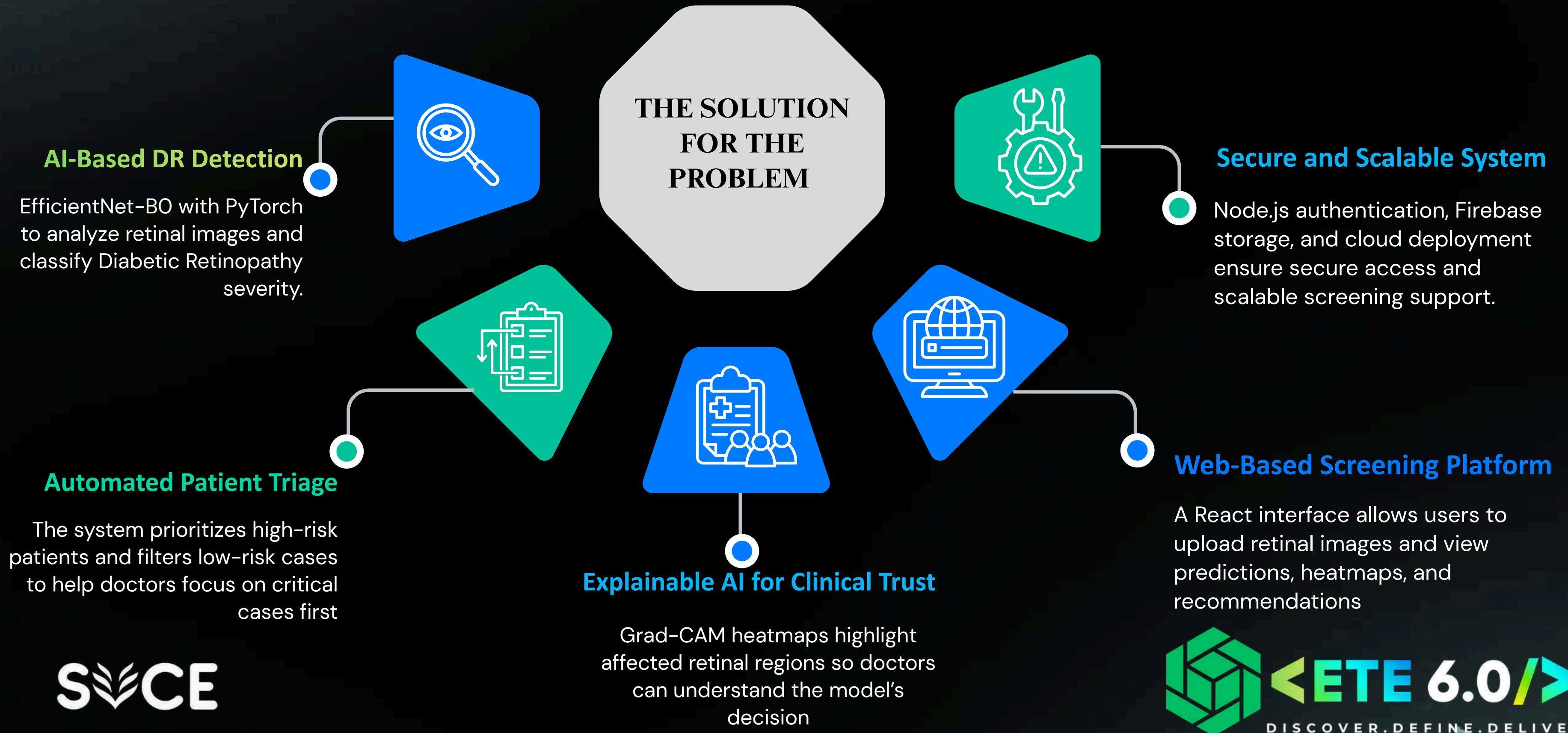
PROBLEM STATEMENT:

AI-Powered Diabetic Retinopathy Screening For Eye Camps

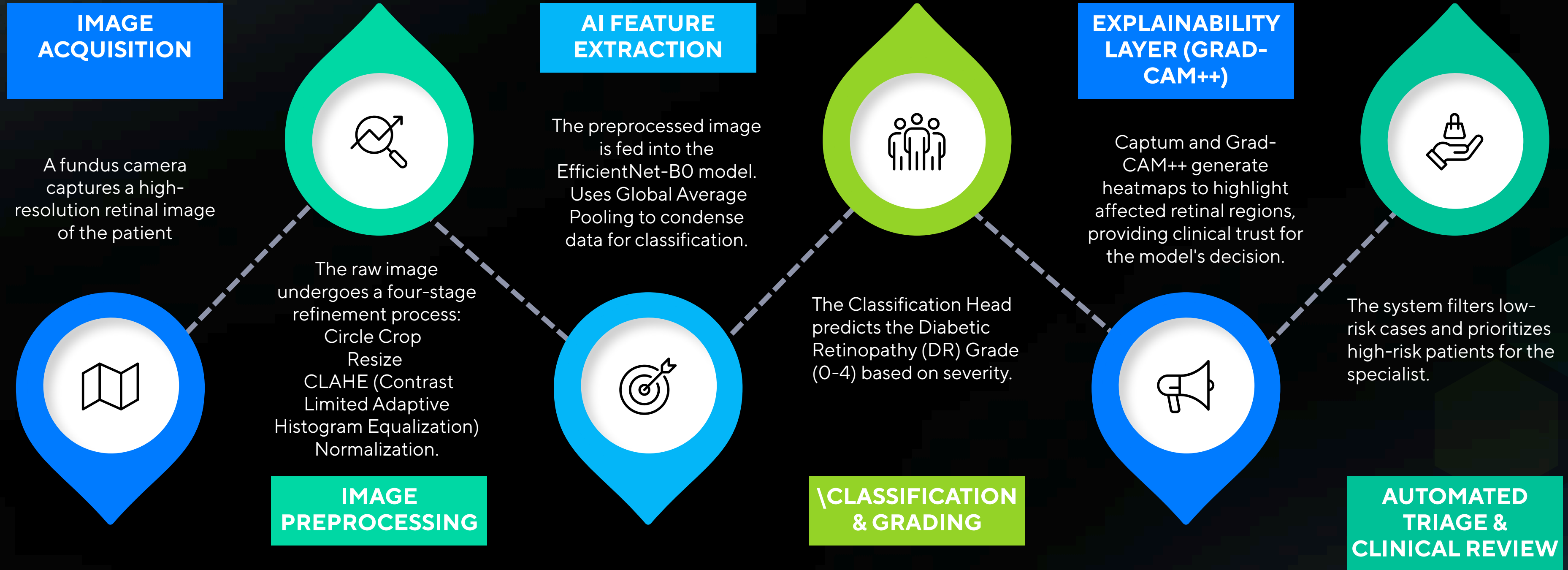
PROBLEM STATEMENT



SOLUTION

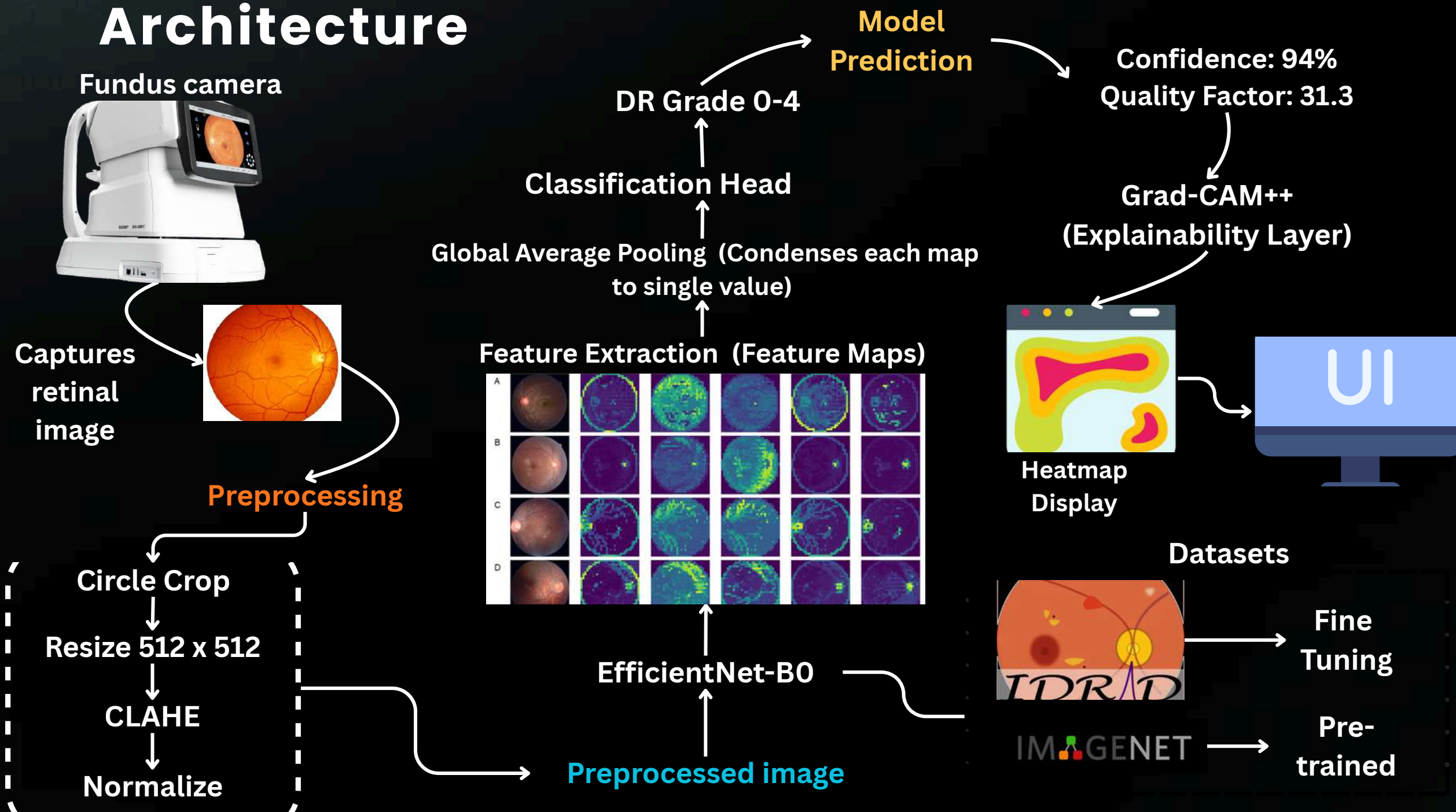


FLOW DIAGRAM



TECHNICAL REQUIREMENTS

Architecture



Tech Stack



NOVELTY / USE CASE

AI-Powered Triage for High-Volume Screening

An intelligent triage assistant, automatically prioritizing high-risk patients in large eye screening camps



Explainable AI for Doctor Trust

provides visual heatmaps showing the retinal regions influencing the prediction, helping doctors verify results.



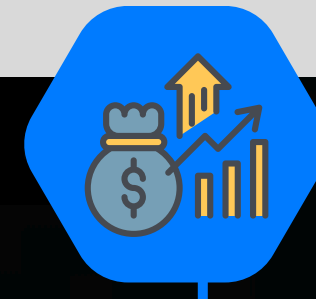
Built for Real-World Field Conditions

The model is trained to handle blurred images, poor lighting, and low contrast, making it suitable for rural screening environments.



Faster and Scalable Screening

By filtering healthy cases automatically, the system helps reduce doctor workload and speed up patient screening



Early Detection to Prevent Blindness

Enables early identification of severe Diabetic Retinopathy, helping ensure timely treatment and preventing avoidable vision loss



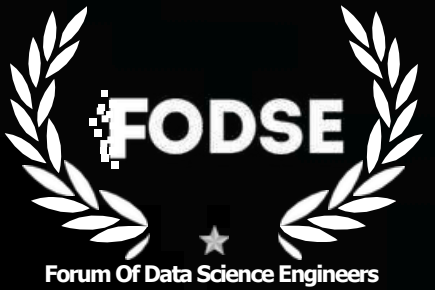
BUSINESS MODEL

- The project is built on the premise of operational and economic feasibility to solve the "Rural Bottleneck"
- Value Proposition: An AI triage system that filters out the 70–80% of "Healthy" or "Mild" cases, allowing specialists to focus on critical patients.
- Resource Efficiency: Reduces specialist fatigue and prevents oversight in high-volume rural camps.
- Cost Management: Using open-source tools keeps development costs low, while early detection reduces long-term healthcare expenses.

TARGET MARKET

- Rural Eye Camps: Specifically targeting camps where a single specialist may have to screen hundreds of patients.
- Low-Resource Environments: Field camps where retinal images often suffer from motion blur, poor lighting, or low contrast.
- Healthcare Providers & NGOs: Organizations needing a scalable solution to automate patient triage and prioritize high-risk cases.
- Clinical Research: Utilizing large-scale public datasets (EyePACS, Messidor, APTOS) for continuous model improvement.

TEAM DETAILS



Name	Branch/Year	College	E-mail	Phone Number
Kesava navya N	CSE-II	Sri Venkateshwara College of Engineering	2024cs0394@svce.ac.in	9445445959
Navya S	CSE-II	Sri Venkateshwara College of Engineering	2024cs0498@svce.ac.in	6380973913
C Dhinesh	CSE-III	Sri Venkateshwara College of Engineering	2023cs0068@svce.ac.in	94453 99983
Deepan K R	CSE-III	Sri Venkateshwara College of Engineering	2023cs0483@svce.ac.in	63826 27877
Balamurugan G	CSE-III	Sri Venkateshwara College of Engineering	2023cs0068@svce.ac.in	8489533066